

Debian Buster from Discovery to Mastery

THE DEBIAN ADMINISTRATOR'S HANDBOOK

Raphaël Hertzog Roland Mas

The Debian Administrator's Handbook

Debian Buster from Discovery to Mastery

Raphaël Hertzog and Roland Mas

Freexian SARL

Sorbiers

The Debian Administrator's Handbook
Raphaël Hertzog and Roland Mas

Copyright © 2003-2020 Raphaël Hertzog

Copyright © 2006-2015 Roland Mas

Copyright © 2012-2020 Freexian SARL

ISBN: 979-10-91414-19-7 (English paperback)

ISBN: 979-10-91414-20-3 (English ebook)

This book is available under the terms of two licenses compatible with the Debian Free Software Guidelines.

Creative Commons License Notice: This book is licensed under a Creative Commons Attribution-ShareAlike 3.0 Unported License.

➨ https://creativecommons.org/licenses/by-sa/3.0/

GNU General Public License Notice: This book is free documentation: you can redistribute it and/or modify it under the terms of the GNU General Public License as published by the Free Software Foundation, either version 2 of the License, or (at your option) any later version.

This book is distributed in the hope that it will be useful, but WITHOUT ANY WARRANTY; without even the implied warranty of MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE. See the GNU General Public License for more details.

You should have received a copy of the GNU General Public License along with this program. If not, see https://www.gnu.org/licenses/.

Show your appreciation



This book is published under a free license because we want everybody to benefit from it. That said maintaining it takes time and lots of effort, and we appreciate being thanked for this. If you find this book valuable, please consider contributing to its continued maintenance either by buying a paperback copy or by making a donation through the book's official website:

➨ https://debian-handbook.info

Table of Contents

1. The Debian Project 1

1.1 What Is Debian? 2

1.1.1 A Multi-Platform Operating System 2

1.1.2 The Quality of Free Software 4

1.1.3 The Legal Framework: A Non-Profit Organization 4

1.2 The Foundation Documents 5

1.2.1 The Commitment towards Users 5

1.2.2 The Debian Free Software Guidelines 6

1.3 The Inner Workings of the Debian Project 9

1.3.1 The Debian Developers 9

1.3.2 The Active Role of Users 13

Reporting bugs 14

Translation and documentation 15

Sending fixes 15

Other ways of contributing 17

1.3.3 Teams and Sub-Projects 17

Existing Debian Sub-Projects 18

Administrative Teams 19

Development Teams, Transversal Teams 20

1.4 Follow Debian News 21

1.5 The Role of Distributions 23

1.5.1 The Installer: `debian-installer` 23

1.5.2 The Software Library 23

1.6 Lifecycle of a Release 24

1.6.1 The *Experimental* Status 24

1.6.2 The *Unstable* Status 24

1.6.3 Migration to *Testing* 26

1.6.4 The Promotion from *Testing* to *Stable* 27

1.6.5 The *Oldstable* and *Oldoldstable* Status 30

2. Presenting the Case Study 33

2.1 Fast Growing IT Needs 34

2.2 Master Plan 34

2.3 Why a GNU/Linux Distribution? 35

2.4 Why the Debian Distribution? 37

2.4.1 Commercial and Community Driven Distributions . . . 37
2.5 Why Debian Buster? . . . 38
3. Analyzing the Existing Setup and Migrating 41
3.1 Coexistence in Heterogeneous Environments . . . 42
3.1.1 Integration with Windows Machines . . . 42
3.1.2 Integration with OS X machines . . . 42
3.1.3 Integration with Other Linux/Unix Machines . . . 42
3.2 How To Migrate . . . 43
3.2.1 Survey and Identify Services . . . 43
Network and Processes . . . 44
3.2.2 Backing up the Configuration . . . 45
3.2.3 Taking Over an Existing Debian Server . . . 45
3.2.4 Installing Debian . . . 46
3.2.5 Installing and Configuring the Selected Services . . . 47
4. Installation 51
4.1 Installation Methods . . . 52
4.1.1 Installing from a CD-ROM/DVD-ROM . . . 52
4.1.2 Booting from a USB Key . . . 53
4.1.3 Installing through Network Booting . . . 54
4.1.4 Other Installation Methods . . . 54
4.2 Installing, Step by Step . . . 54
4.2.1 Booting and Starting the Installer . . . 54
4.2.2 Selecting the language . . . 56
4.2.3 Selecting the country . . . 57
4.2.4 Selecting the keyboard layout . . . 57
4.2.5 Detecting Hardware . . . 58
4.2.6 Loading Components . . . 58
4.2.7 Detecting Network Hardware . . . 58
4.2.8 Configuring the Network . . . 59
4.2.9 Administrator Password . . . 59
4.2.10 Creating the First User . . . 60
4.2.11 Configuring the Clock . . . 61
4.2.12 Detecting Disks and Other Devices . . . 61
4.2.13 Starting the Partitioning Tool . . . 61
Guided partitioning . . . 63
Manual Partitioning . . . 65
Configuring Multidisk Devices (Software RAID) . . . 67
Configuring the Logical Volume Manager (LVM) . . . 67
Setting Up Encrypted Partitions . . . 68
4.2.14 Installing the Base System . . . 69
4.2.15 Configuring the Package Manager (`apt`) . . . 69
4.2.16 Debian Package Popularity Contest . . . 70
4.2.17 Selecting Packages for Installation . . . 71

4.2.18 Installing the GRUB Bootloader . . . 71
4.2.19 Finishing the Installation and Rebooting . . . 72
4.3 After the First Boot . . . 73
4.3.1 Installing Additional Software . . . 73
4.3.2 Upgrading the System . . . 74
5. Packaging System: Tools and Fundamental Principles 77
5.1 Structure of a Binary Package . . . 78
5.2 Package Meta-Information . . . 80
5.2.1 Description: the `control` File . . . 80
Dependencies: the Depends Field . . . 81
Conflicts: the Conflicts field . . . 83
Incompatibilities: the Breaks Field . . . 83
Provided Items: the Provides Field . . . 83
Replacing Files: The Replaces Field . . . 86
5.2.2 Configuration Scripts . . . 86
Installation and Upgrade . . . 87
Package Removal . . . 87
5.2.3 Checksums, List of Configuration Files . . . 89
5.3 Structure of a Source Package . . . 90
5.3.1 Format . . . 90
5.3.2 Usage within Debian . . . 93
5.4 Manipulating Packages with `dpkg` . . . 94
5.4.1 Installing Packages . . . 94
5.4.2 Package Removal . . . 96
5.4.3 Querying `dpkg`'s Database and Inspecting `.deb` Files . . . 96
5.4.4 `dpkg`'s Log File . . . 101
5.4.5 Multi-Arch Support . . . 101
Enabling Multi-Arch . . . 101
Multi-Arch Related Changes . . . 102
5.5 Coexistence with Other Packaging Systems . . . 103
6. Maintenance and Updates: The APT Tools 107
6.1 Filling in the `sources.list` File . . . 108
6.1.1 Syntax . . . 108
6.1.2 Repositories for *Stable* Users . . . 110
Security Updates . . . 111
Stable Updates . . . 111
Proposed Updates . . . 112
Stable Backports . . . 112
6.1.3 Repositories for *Testing*/*Unstable* Users . . . 112
The Experimental *Repository* . . . 113
6.1.4 Using Alternate Mirrors . . . 114
6.1.5 Non-Official Resources: mentors.debian.net . . . 114
6.1.6 Caching Proxy for Debian Packages . . . 115

6.2 aptitude, apt-get, and apt Commands . . . 116
6.2.1 Initialization . . . 116
6.2.2 Installing and Removing . . . 117
6.2.3 System Upgrade . . . 120
6.2.4 Configuration Options . . . 120
6.2.5 Managing Package Priorities . . . 121
6.2.6 Working with Several Distributions . . . 124
6.2.7 Tracking Automatically Installed Packages . . . 125
6.3 The apt-cache Command . . . 126
6.4 The apt-file Command . . . 128
6.5 Frontends: aptitude, synaptic . . . 128
6.5.1 aptitude . . . 128
Managing Recommendations, Suggestions and Tasks . . . 130
Better Solver Algorithms . . . 131
6.5.2 synaptic . . . 131
6.6 Checking Package Authenticity . . . 132
6.7 Upgrading from One Stable Distribution to the Next . . . 134
6.7.1 Recommended Procedure . . . 134
6.7.2 Handling Problems after an Upgrade . . . 135
6.7.3 Cleaning Up after an Upgrade . . . 136
Packages removed from the Debian Archive . . . 137
Dummy and Transitional Packages . . . 137
Old or Unused Configuration Files . . . 137
Files not owned by any Package . . . 138
6.8 Keeping a System Up to Date . . . 138
6.9 Automatic Upgrades . . . 140
6.9.1 Configuring dpkg . . . 140
6.9.2 Configuring APT . . . 141
6.9.3 Configuring debconf . . . 141
6.9.4 Handling Command Line Interactions . . . 141
6.9.5 The Miracle Combination . . . 141
6.10 Searching for Packages . . . 142
7. Solving Problems and Finding Relevant Information 147
7.1 Documentation Sources . . . 148
7.1.1 Manual Pages . . . 148
7.1.2 *info* Documents . . . 150
7.1.3 Specific Documentation . . . 151
7.1.4 Websites . . . 151
7.1.5 Tutorials (*HOWTO*) . . . 152
7.2 Common Procedures . . . 153
7.2.1 Configuring a Program . . . 153
7.2.2 Monitoring What Daemons Are Doing . . . 154
7.2.3 Asking for Help on a Mailing List . . . 155

7.2.4 Reporting a Bug When a Problem Is Too Difficult . . . 156

8. Basic Configuration: Network, Accounts, Printing... 159

8.1 Configuring the System for Another Language . . . 160
8.1.1 Setting the Default Language . . . 160
8.1.2 Configuring the Keyboard . . . 161
8.1.3 Migrating to UTF-8 . . . 162
8.2 Configuring the Network . . . 163
8.2.1 Ethernet Interface . . . 165
8.2.2 Wireless Interface . . . 166
Installing the required firmwares . . . 166
Wireless specific entries in `/etc/network/interfaces` . . . 167
8.2.3 Connecting with PPP through a PSTN Modem . . . 167
8.2.4 Connecting through an ADSL Modem . . . 168
Modems Supporting PPPOE . . . 168
Modems Supporting PPTP . . . 169
Modems Supporting DHCP . . . 169
8.2.5 Automatic Network Configuration for Roaming Users . . . 169
8.3 Setting the Hostname and Configuring the Name Service . . . 170
8.3.1 Name Resolution . . . 171
Configuring DNS Servers . . . 171
The `/etc/hosts` file . . . 171
8.4 User and Group Databases . . . 172
8.4.1 User List: `/etc/passwd` . . . 173
8.4.2 The Hidden and Encrypted Password File: `/etc/shadow` . . . 173
8.4.3 Modifying an Existing Account or Password . . . 174
8.4.4 Disabling an Account . . . 174
8.4.5 Group List: `/etc/group` . . . 174
8.5 Creating Accounts . . . 175
8.6 Shell Environment . . . 176
8.7 Printer Configuration . . . 178
8.8 Configuring the Bootloader . . . 179
8.8.1 Identifying the Disks . . . 179
8.8.2 Configuring LILO . . . 181
8.8.3 GRUB 2 Configuration . . . 182
8.9 Other Configurations: Time Synchronization, Logs, Sharing Access... . . . 183
8.9.1 Timezone . . . 183
8.9.2 Time Synchronization . . . 184
For Workstations . . . 185
For Servers . . . 185
8.9.3 Rotating Log Files . . . 186
8.9.4 Sharing Administrator Rights . . . 186
8.9.5 List of Mount Points . . . 186
8.9.6 `locate` and `updatedb` . . . 189

8.10 Compiling a Kernel . . . 189
8.10.1 Introduction and Prerequisites . . . 189
8.10.2 Getting the Sources . . . 190
8.10.3 Configuring the Kernel . . . 191
8.10.4 Compiling and Building the Package . . . 192
8.10.5 Compiling External Modules . . . 192
8.10.6 Applying a Kernel Patch . . . 193
8.11 Installing a Kernel . . . 194
8.11.1 Features of a Debian Kernel Package . . . 194
8.11.2 Installing with dpkg . . . 195
9. Unix Services 197
9.1 System Boot . . . 198
9.1.1 The systemd init system . . . 198
9.1.2 The System V init system . . . 204
9.2 Remote Login . . . 207
9.2.1 Secure Remote Login: SSH . . . 207
Key-Based Authentication . . . 208
Using Remote X11 Applications . . . 210
Creating Encrypted Tunnels with Port Forwarding . . . 210
9.2.2 Using Remote Graphical Desktops . . . 212
9.3 Managing Rights . . . 213
9.4 Administration Interfaces . . . 215
9.4.1 Administrating on a Web Interface: webmin . . . 216
9.4.2 Configuring Packages: debconf . . . 217
9.5 syslog System Events . . . 218
9.5.1 Principle and Mechanism . . . 218
9.5.2 The Configuration File . . . 219
Syntax of the Selector . . . 219
Syntax of Actions . . . 219
9.6 The inetd Super-Server . . . 220
9.7 Scheduling Tasks with cron and atd . . . 222
9.7.1 Format of a crontab File . . . 222
9.7.2 Using the at Command . . . 224
9.8 Scheduling Asynchronous Tasks: anacron . . . 225
9.9 Quotas . . . 226
9.10 Backup . . . 227
9.10.1 Backing Up with rsync . . . 227
9.10.2 Restoring Machines without Backups . . . 229
9.11 Hot Plugging: *hotplug* . . . 230
9.11.1 Introduction . . . 230
9.11.2 The Naming Problem . . . 230
9.11.3 How *udev* Works . . . 231
9.11.4 A concrete example . . . 232

9.12 Power Management: Advanced Configuration and Power Interface (ACPI) 234
10. Network Infrastructure 237
10.1 Gateway 238
10.2 X.509 certificates 240
10.2.1 Creating gratis trusted certificates 240
10.2.2 Public Key Infrastructure: *easy-rsa* 243
10.3 Virtual Private Network 247
10.3.1 OpenVPN 247
Configuring the OpenVPN Server 248
Configuring the OpenVPN Server 248
Configuring the OpenVPN Client 249
10.3.2 Virtual Private Network with SSH 249
10.3.3 IPsec 250
10.3.4 PPTP 250
Configuring the Client 251
Configuring the Server 252
10.4 Quality of Service 254
10.4.1 Principle and Mechanism 254
10.4.2 Configuring and Implementing 255
Reducing Latencies: `wondershaper` 255
Standard Configuration 256
10.5 Dynamic Routing 256
10.6 IPv6 257
10.6.1 Tunneling 258
10.7 Domain Name Servers (DNS) 259
10.7.1 DNS software 260
10.7.2 Configuring bind 260
10.8 DHCP 263
10.8.1 Configuring 263
10.8.2 DHCP and DNS 264
10.9 Network Diagnosis Tools 264
10.9.1 Local Diagnosis: `netstat` 265
10.9.2 Remote Diagnosis: `nmap` 266
10.9.3 Sniffers: `tcpdump` and `wireshark` 267
11. Network Services: Postfix, Apache, NFS, Samba, Squid, LDAP, SIP, XMPP, TURN 271
11.1 Mail Server 272
11.1.1 Installing Postfix 272
11.1.2 Configuring Virtual Domains 275
Virtual Alias Domains 276
Virtual Mailbox Domains 276
11.1.3 Restrictions for Receiving and Sending 277
IP-Based Access Restrictions 278

Checking the Validity of the EHLO or HELO Commands . . . 279
Accepting or Refusing Based on the Announced Sender . . . 280
Accepting or Refusing Based on the Recipient . . . 281
Restrictions Associated with the DATA Command . . . 281
Applying Restrictions . . . 282
Filtering Based on the Message Contents . . . 282
11.1.4 Setting Up *greylisting* . . . 283
11.1.5 Customizing Filters Based On the Recipient . . . 285
11.1.6 Integrating an Antivirus . . . 286
11.1.7 Fighting Spam with SPF, DKIM and DMARC . . . 287
Integrating the Sender Policy Framework (SPF) . . . 287
Integrating DomainKeys (DKIM) Signing and Checking . . . 288
Integrating Domain-based Message Authentication, Reporting and Conformance (DMARC) . . . 290
11.1.8 Authenticated SMTP . . . 291
11.2 Web Server (HTTP) . . . 293
11.2.1 Installing Apache . . . 293
11.2.2 Adding support for SSL . . . 294
11.2.3 Configuring Virtual Hosts . . . 295
11.2.4 Common Directives . . . 296
Requiring Authentication . . . 297
Restricting Access . . . 298
11.2.5 Log Analyzers . . . 299
11.3 FTP File Server . . . 301
11.4 NFS File Server . . . 302
11.4.1 Securing NFS . . . 302
11.4.2 NFS Server . . . 303
11.4.3 NFS Client . . . 304
11.5 Setting Up Windows Shares with Samba . . . 305
11.5.1 Samba Server . . . 305
Configuring with `debconf` . . . 305
Configuring Manually . . . 306
11.5.2 Samba Client . . . 307
The `smbclient` Program . . . 307
Mounting Windows Shares . . . 307
Printing on a Shared Printer . . . 308
11.6 HTTP/FTP Proxy . . . 308
11.6.1 Installing . . . 309
11.6.2 Configuring a Cache . . . 309
11.6.3 Configuring a Filter . . . 310
11.7 LDAP Directory . . . 310
11.7.1 Installing . . . 311
11.7.2 Filling in the Directory . . . 312
11.7.3 Managing Accounts with LDAP . . . 313

Configuring NSS . . . 313
Configuring PAM . . . 315
Securing LDAP Data Exchanges . . . 316
11.8 Real-Time Communication Services . . . 319
11.8.1 DNS settings for RTC services . . . 320
11.8.2 TURN Server . . . 320
11.8.3 SIP Proxy Server . . . 321
Install the SIP proxy . . . 321
11.8.4 XMPP Server . . . 322
Install the XMPP server . . . 322
Managing the XMPP server . . . 323
11.8.5 Running services on port 443 . . . 323
11.8.6 Adding WebRTC . . . 324
12. Advanced Administration 327
12.1 RAID and LVM . . . 328
12.1.1 Software RAID . . . 328
Different RAID Levels . . . 329
Setting up RAID . . . 331
Backing up the Configuration . . . 337
12.1.2 LVM . . . 339
LVM Concepts . . . 339
Setting up LVM . . . 340
LVM Over Time . . . 344
12.1.3 RAID or LVM? . . . 346
12.2 Virtualization . . . 349
12.2.1 Xen . . . 350
12.2.2 LXC . . . 356
Preliminary Steps . . . 356
Network Configuration . . . 357
Setting Up the System . . . 358
Starting the Container . . . 359
12.2.3 Virtualization with KVM . . . 360
Preliminary Steps . . . 361
Network Configuration . . . 361
Installation with `virt-install` . . . 361
Managing Machines with `virsh` . . . 363
Installing an RPM based system in Debian with yum . . . 364
12.3 Automated Installation . . . 365
12.3.1 Fully Automatic Installer (FAI) . . . 366
12.3.2 Preseeding Debian-Installer . . . 367
Using a Preseed File . . . 367
Creating a Preseed File . . . 368
Creating a Customized Boot Media . . . 368

12.3.3 Simple-CDD: The All-In-One Solution . . . 370
Creating Profiles . . . 370
Configuring and Using `build-simple-cdd` . . . 371
Generating an ISO Image . . . 371
12.4 Monitoring . . . 371
12.4.1 Setting Up Munin . . . 372
Configuring Hosts To Monitor . . . 372
Configuring the Grapher . . . 374
12.4.2 Setting Up Nagios . . . 374
Installing . . . 375
Configuring . . . 375
13. Workstation 381
13.1 Configuring the X11 Server . . . 382
13.2 Customizing the Graphical Interface . . . 383
13.2.1 Choosing a Display Manager . . . 383
13.2.2 Choosing a Window Manager . . . 383
13.2.3 Menu Management . . . 384
13.3 Graphical Desktops . . . 385
13.3.1 GNOME . . . 385
13.3.2 KDE and Plasma . . . 386
13.3.3 Xfce and Others . . . 387
13.3.4 Other Desktop Environments . . . 388
13.4 Email . . . 389
13.4.1 Evolution . . . 389
13.4.2 KMail . . . 390
13.4.3 Thunderbird . . . 390
13.5 Web Browsers . . . 391
13.6 Development . . . 393
13.6.1 Tools for GTK+ on GNOME . . . 393
13.6.2 Tools for Qt . . . 393
13.7 Collaborative Work . . . 394
13.7.1 Working in Groups: *groupware* . . . 394
13.7.2 Collaborative Work With FusionForge . . . 394
13.8 Office Suites . . . 395
13.9 Emulating Windows: Wine . . . 396
13.10 Real-Time Communications software . . . 397
14. Security 401
14.1 Defining a Security Policy . . . 402
14.2 Firewall or Packet Filtering . . . 403
14.2.1 nftables Behavior . . . 404
14.2.2 Moving from iptables to nftables . . . 406
14.2.3 Syntax of `nft` . . . 408
14.2.4 Installing the Rules at Each Boot . . . 409

14.3 Supervision: Prevention, Detection, Deterrence . . . 410
14.3.1 Monitoring Logs with `logcheck` . . . 410
14.3.2 Monitoring Activity . . . 411
In Real Time . . . 411
History . . . 411
14.3.3 Avoiding Intrusion . . . 412
14.3.4 Detecting Changes . . . 413
Auditing Packages with `dpkg --verify` . . . 413
Auditing Packages: `debsums` and its Limits . . . 414
Monitoring Files: AIDE . . . 414
14.3.5 Detecting Intrusion (IDS/NIDS) . . . 416
14.4 Introduction to AppArmor . . . 417
14.4.1 Principles . . . 417
14.4.2 Enabling AppArmor and managing AppArmor profiles . . . 417
14.4.3 Creating a new profile . . . 418
14.5 Introduction to SELinux . . . 424
14.5.1 Principles . . . 424
14.5.2 Setting Up SELinux . . . 426
14.5.3 Managing an SELinux System . . . 427
Managing SELinux Modules . . . 427
Managing Identities . . . 428
Managing File Contexts, Ports and Booleans . . . 429
14.5.4 Adapting the Rules . . . 430
Writing a `.fc` file . . . 430
Writing a `.if` File . . . 430
Writing a `.te` File . . . 432
Compiling the Files . . . 435
14.6 Other Security-Related Considerations . . . 435
14.6.1 Inherent Risks of Web Applications . . . 435
14.6.2 Knowing What To Expect . . . 436
14.6.3 Choosing the Software Wisely . . . 437
14.6.4 Managing a Machine as a Whole . . . 438
14.6.5 Users Are Players . . . 438
14.6.6 Physical Security . . . 439
14.6.7 Legal Liability . . . 439
14.7 Dealing with a Compromised Machine . . . 440
14.7.1 Detecting and Seeing the Cracker's Intrusion . . . 440
14.7.2 Putting the Server Off-Line . . . 440
14.7.3 Keeping Everything that Could Be Used as Evidence . . . 441
14.7.4 Re-installing . . . 442
14.7.5 Forensic Analysis . . . 442
14.7.6 Reconstituting the Attack Scenario . . . 443
15. Creating a Debian Package 447

15.1 Rebuilding a Package from its Sources . . . 448
15.1.1 Getting the Sources . . . 448
15.1.2 Making Changes . . . 448
15.1.3 Starting the Rebuild . . . 450
15.2 Building your First Package . . . 451
15.2.1 Meta-Packages or Fake Packages . . . 451
15.2.2 Simple File Archive . . . 452
15.3 Creating a Package Repository for APT . . . 456
15.4 Becoming a Package Maintainer . . . 458
15.4.1 Learning to Make Packages . . . 458
Rules . . . 458
Procedures . . . 459
Tools . . . 459
15.4.2 Acceptance Process . . . 461
Prerequisites . . . 461
Registration . . . 461
Accepting the Principles . . . 462
Checking Skills . . . 462
Final Approval . . . 463

16. Conclusion: Debian's Future 465
16.1 Upcoming Developments . . . 466
16.2 Debian's Future . . . 466
16.3 Future of this Book . . . 467

A. Derivative Distributions 469
A.1 Census and Cooperation . . . 469
A.2 Ubuntu . . . 469
A.3 Linux Mint . . . 470
A.4 Knoppix . . . 471
A.5 Aptosid and Siduction . . . 471
A.6 Grml . . . 472
A.7 Tails . . . 472
A.8 Kali Linux . . . 472
A.9 Devuan . . . 472
A.10 DoudouLinux . . . 472
A.11 Raspbian . . . 473
A.12 PureOS . . . 473
A.13 SteamOS . . . 473
A.14 And Many More . . . 473

B. Short Remedial Course 475
B.1 Shell and Basic Commands . . . 475
B.1.1 Browsing the Directory Tree and Managing Files . . . 475
B.1.2 Displaying and Modifying Text Files . . . 476

B.1.3 Searching for Files and within Files . . . 477
B.1.4 Managing Processes . . . 477
B.1.5 System Information: Memory, Disk Space, Identity . . . 477
B.2 Organization of the Filesystem Hierarchy . . . 478
B.2.1 The Root Directory . . . 478
B.2.2 The User's Home Directory . . . 479
B.3 Inner Workings of a Computer: the Different Layers Involved . . . 480
B.3.1 The Deepest Layer: the Hardware . . . 480
B.3.2 The Starter: the BIOS or UEFI . . . 481
B.3.3 The Kernel . . . 482
B.3.4 The User Space . . . 482
B.4 Some Tasks Handled by the Kernel . . . 482
B.4.1 Driving the Hardware . . . 482
B.4.2 Filesystems . . . 483
B.4.3 Shared Functions . . . 484
B.4.4 Managing Processes . . . 484
B.4.5 Rights Management . . . 485
B.5 The User Space . . . 485
B.5.1 Process . . . 486
B.5.2 Daemons . . . 486
B.5.3 Inter-Process Communications . . . 487
B.5.4 Libraries . . . 488
Index 489

Preface

I'm pleased to have this opportunity to welcome you to Debian and the Debian Administrator's Handbook. Many people have chosen Debian: around 10% of the web servers on the Internet run Debian. When you include operating systems based on Debian, this number is closer to 20%. Debian was selected as the operating system of choice for the International Space Station. Whether it is cutting edge physics research or a project to help grow food while fighting pollution, Debian has been used to power the computers that make it possible.

Why does Debian have appeal across large corporations, researchers, activists and hobbyists? I think that the answer lies in Debian's flexibility and community.

Debian is flexible. Yes, it provides an excellent general-purpose operating system out of the box. It also provides the tools to customize Debian to whatever environment you find yourself working in. Whether it is a cloud and container architecture, a large collection of workstations, individual computers, or an appliance, Debian provides the flexibility to work well in that environment. You will find the tools and examples you need to meet your needs.

The Debian community is a meeting place for diverse individuals and interests: developers from the largest corporations work alongside volunteers, researchers, and users. Whether it is security experts, web developers, systems programmers or architects, we are all represented. You can be part of this community. When you find ways that Debian can be better, we welcome your contribution.

We come together to produce a world-class free operating system. No one company controls Debian; no one agenda defines our work. Instead, each of us has the power to improve Debian in the ways that matter to us. Thank you for taking a look at what we've built. I hope you like it.

This book is an excellent way to explore Debian. I've been recommending it to friends for years when they wanted to learn more about Debian, and I am pleased to have the opportunity to recommend it more widely. This handbook is written and maintained by long-standing members of the Debian community. Some of the same people who are working to develop the operating system have joined together to help you understand it. And of course the book is developed using a community process similar to Debian itself with the same emphasis on freedom.

August 2019

Sam Hartman (Debian Project Leader)

Foreword

Linux has been garnering strength for a number of years now, and its growing popularity drives more and more users to make the jump. The first step on that path is to pick a distribution. This is an important decision, because each distribution has its own peculiarities, and future migration costs can be avoided if the right choice is made from the start.

BACK TO BASICS

Linux distribution, Linux kernel

Strictly speaking, Linux is only a kernel, the core piece of software which sits between the hardware and the applications.

A "Linux distribution" is a full operating system; it usually includes the Linux kernel, an installer program, and most importantly applications and other software required to turn a computer into a tool that is actually useful.

Debian GNU/Linux is a "generic" Linux distribution that fits most users. The purpose of this book is to show its many aspects so that you can make an informed decision when choosing.

Why This Book?

CULTURE

Commercial distributions

Most Linux distributions are backed by a for-profit company that develops them and sells them under some kind of commercial scheme. Examples include *Ubuntu*, mainly developed by *Canonical Ltd.*; *Red Hat Enterprise Linux*, by *Red Hat*; and *SUSE Linux*, maintained and made commercially available by *Novell*.

At the other end of the spectrum lie the likes of Debian and the Apache Software Foundation (which hosts the development for the Apache web server). Debian is above all a project in the Free Software world, implemented by volunteers working together through the Internet. While some of them do work on Debian as part of their paid job in various companies, the project as a whole is not attached to any company in particular, nor does any one company have a greater say in the project's affairs than what purely volunteer contributors have.

Linux has gathered a fair amount of media coverage over the years; it mostly benefits the distributions supported by a real marketing department — in other words, company-backed distributions (Ubuntu, Red Hat, SUSE, and so on). But Debian is far from being a marginal distribution; multiple studies have shown over the years that it is widely used both on servers and on desktops. This is particularly true among web servers where Debian and Ubuntu are the leading Linux distributions.

➨ https://w3techs.com/technologies/details/os-linux/all/all

The purpose of this book is to help you discover this distribution. We hope to share the experience that we have gathered since we joined the project as developers and contributors in 1998 (Raphaël) and 2000 (Roland). With any luck, our enthusiasm will be communicative, and maybe you will join us sometime...

The first edition of this book (in 2004) served to fill a gaping hole: it was the first French-language book that focused exclusively on Debian. At that time, many other books were written on the topic both for French-speaking and English-speaking readers. Unfortunately almost none of them got updated, and over the years the situation slipped back to one where there were very few good books on Debian. We hope that this book, which has started a new life with its translation into English (and several translations from English into various other languages), will fill this gap and help many users.

Who Is this Book For?

We tried to make this book useful for many categories of readers. First, systems administrators (both beginners and experienced) will find explanations about the installation and deployment of Debian on many computers. They will also get a glimpse of most of the services available on Debian, along with matching configuration instructions and a description of the specifics coming from the distribution. Understanding the mechanisms involved in Debian's development will enable them to deal with unforeseen problems, knowing that they can always find help within the community.

Users of another Linux distribution, or of another Unix variant, will discover the specifics of Debian, and should become operational very quickly while benefiting fully from the unique advantages of this distribution.

Finally, readers who already have some knowledge of Debian and want to know more about the community behind it should see their expectations fulfilled. This book should make them much closer to joining us as contributors.

General Approach

All of the generic documentation you can find about GNU/Linux also applies to Debian, since Debian includes most common free software. However, the distribution brings many enhancements, which is why we chose to primarily describe the "Debian way" of doing things.

It is interesting to follow the Debian recommendations, but it is even better to understand their rationale. Therefore, we won't restrict ourselves to practical explanations only; we will also describe the project's workings, so as to provide you with comprehensive and consistent knowledge.

Book Structure

This book is built around a case study providing both support and illustration for all topics being addressed.

NOTE

Web site, authors' email

This book has its own website, which hosts whatever elements that can make it more useful. In particular, it includes an online version of the book with clickable links, and possible errata. Feel free to browse it and to leave us some feedback. We will be happy to read your comments or support messages. Send them by email to hertzog@debian.org (Raphaël) and lolando@debian.org (Roland).

➨ https://debian-handbook.info/

Chapter 1 focuses on a non-technical presentation of the Debian project and describes its goals and organization. These aspects are important because they define a general framework that other chapters will complete with more concrete information.

Chapters 2 and 3 provide a broad outline of the case study. At this point, novice readers can take the time to read **appendix B**, where they will find a short remedial course explaining a number of basic computing notions, as well as concepts inherent to any Unix system.

To get on with our real subject matter, we will quite naturally start with the installation process (**chapter 4**); **chapters 5 and 6** will unveil basic tools that any Debian administrator will use, such as those of the **APT** family, which is largely responsible for the distribution's excellent reputation. These chapters are in no way restricted to professionals, since everyone is their own administrator at home.

Chapter 7 will be an important parenthesis; it describes workflows to efficiently use documentation and to quickly gain an understanding of problems in order to solve them.

The next chapters will be a more detailed tour of the system, starting with basic infrastructure and services (**chapters 8 to 10**) and going progressively up the stack to reach the user applications in **chapter 13**. **Chapter 12** deals with more advanced subjects that will most directly concern administrators of large sets of computers (including servers), while **chapter 14** is a brief introduction to the wider subject of computer security and gives a few keys to avoid most problems.

Chapter 15 is for administrators who want to go further and create their own Debian packages.

VOCABULARY

Debian package

A Debian package is an archive containing all the files required to install a piece of software. It is generally a file with a `.deb` extension, and it can be handled with the `dpkg` command. Also called a *binary package*, it contains files that can be directly used (such as programs or documentation). On the other hand, a *source package* contains the source code for the software and the instructions required for building the binary package.

The present version is already the ninth edition of the book (we include the first four that were only available in French). This edition covers version 10 of Debian, code-named *Buster*. Among

the changes, Debian now supports UEFI Secure Boot, providing some extra safety against attacks on the boot infrastructure, and making it easier to install Debian on new computers where Secure Boot is usually enabled by default. Again at the security level, AppArmor, a Mandatory Access Control system that regulates what various applications are allowed to perform, is now enabled by default. All included packages have obviously been updated, including the GNOME desktop, which is now in its version 3.30.

We have added some notes and remarks in sidebars. They have a variety of roles: they can draw attention to a difficult point, complete a notion of the case study, define some terms, or serve as reminders. Here is a list of the most common of these sidebars:

- BACK TO BASICS: a reminder of some information that is supposed to be known;
- VOCABULARY: defines a technical term, sometimes Debian specific;
- COMMUNITY: highlights important persons or roles within the project;
- POLICY: a rule or recommendation from the Debian Policy. This document is essential within the project, and describes how to package software. The parts of the policy highlighted in this book bring direct benefits to users (for example, knowing that the policy standardizes the location of documentation and examples makes it easy to find them even in a new package).
- TOOL: presents a relevant tool or service;
- IN PRACTICE: theory and practice do not always match; these sidebars contain advice resulting from our experience. They can also give detailed and concrete examples;
- other more or less frequent sidebars are rather explicit: CULTURE, TIP, CAUTION, GOING FURTHER, SECURITY, and so on.

Contributing

This book is developed like a free software project, your input and help is welcome. The most obvious way to contribute is to help translate it into your native language. But that is not the only possibility. You can open bug reports to let us know of mistakes, typos, outdated information, or topics that we should really cover. Or you can submit a merge request with your fix for whatever issue that you identified.

All the instructions to contribute to the book are documented on the book's website:

➨ https://debian-handbook.info/contribute/

Acknowledgments

A Bit of History

In 2003, Nat Makarévitch contacted Raphaël because he wanted to publish a book on Debian in the *Cahier de l'Admin* (Admin's Handbook) collection that he was managing for Eyrolles, a leading

French editor of technical books. Raphaël immediately accepted to write it. The first edition came out on 14th October 2004 and was a huge success — it was sold out barely four months later.

Since then, we have released 7 other editions of the French book, one for each subsequent Debian release (except for Debian 9). Roland, who started working on the book as a proofreader, gradually became its co-author.

While we were obviously satisfied with the book's success, we always hoped that Eyrolles would convince an international editor to translate it into English. We had received numerous comments explaining how the book helped people to get started with Debian, and we were keen to have the book benefit more people in the same way.

Alas, no English-speaking editor that we contacted was willing to take the risk of translating and publishing the book. Not put off by this small setback, we negotiated with our French editor Eyrolles and got back the necessary rights to translate the book into English and publish it ourselves. Thanks to a successful crowdfunding campaign[1], we worked on the translation between December 2011 and May 2012. The "Debian Administrator's Handbook" was born and it was published under a free-software license!

While this was an important milestone, we already knew that the story would not be over for us until we could contribute the French book as an official translation of the English book. This was not possible at that time because the French book was still distributed commercially under a non-free license by Eyrolles.

In 2013, the release of Debian 7 gave us a good opportunity to discuss a new contract with Eyrolles. We convinced them that a license more in line with the Debian values would contribute to the book's success. That wasn't an easy deal to make, and we agreed to setup another crowdfunding campaign[2] to cover some of the costs and reduce the risks involved. The operation was again a huge success and in July 2013, we added a French translation to the Debian Administrator's Handbook.

We would like to thank everybody who contributed to these fundraising campaigns, either by pledging some money or by passing the word around. We couldn't have done it without you.

To save some paper, 5 years after the fundraising campaigns and after two subsequent editions, we dropped the list of persons who opted to be rewarded with a mention of their name in the book. But their names are engraved in the acknowledgments of the Wheezy edition of the book:

➨ https://debian-handbook.info/browse/wheezy/sect.acknowledgments.html

Special Thanks to Contributors

This book would not be what it is without the contributions of several persons who each played an important role during the translation phase and beyond. We would like to thank Marilyne Brun, who helped us to translate the sample chapter and who worked with us to define some

[1] https://www.ulule.com/debian-handbook/

[2] https://www.ulule.com/liberation-cahier-admin-debian/